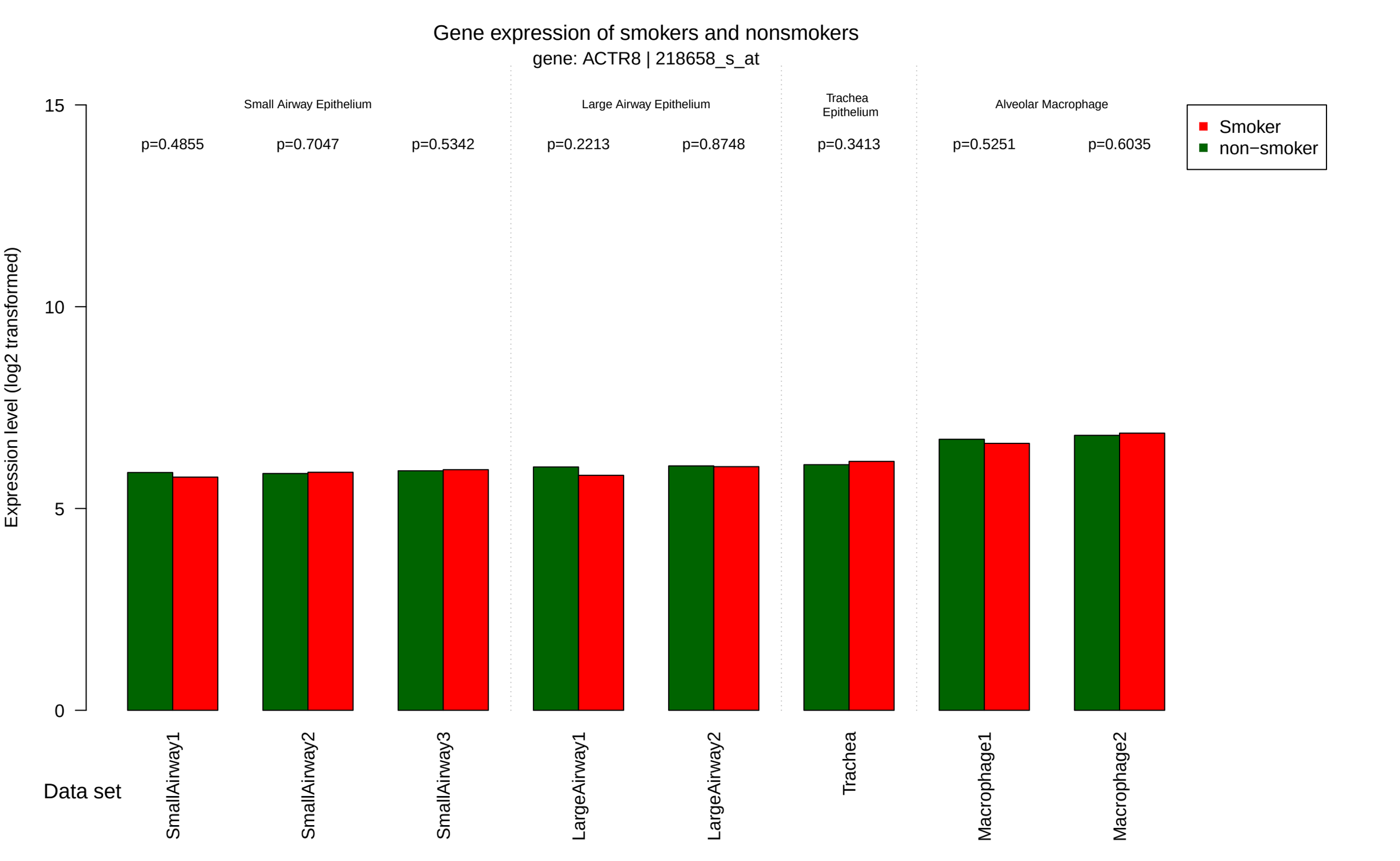
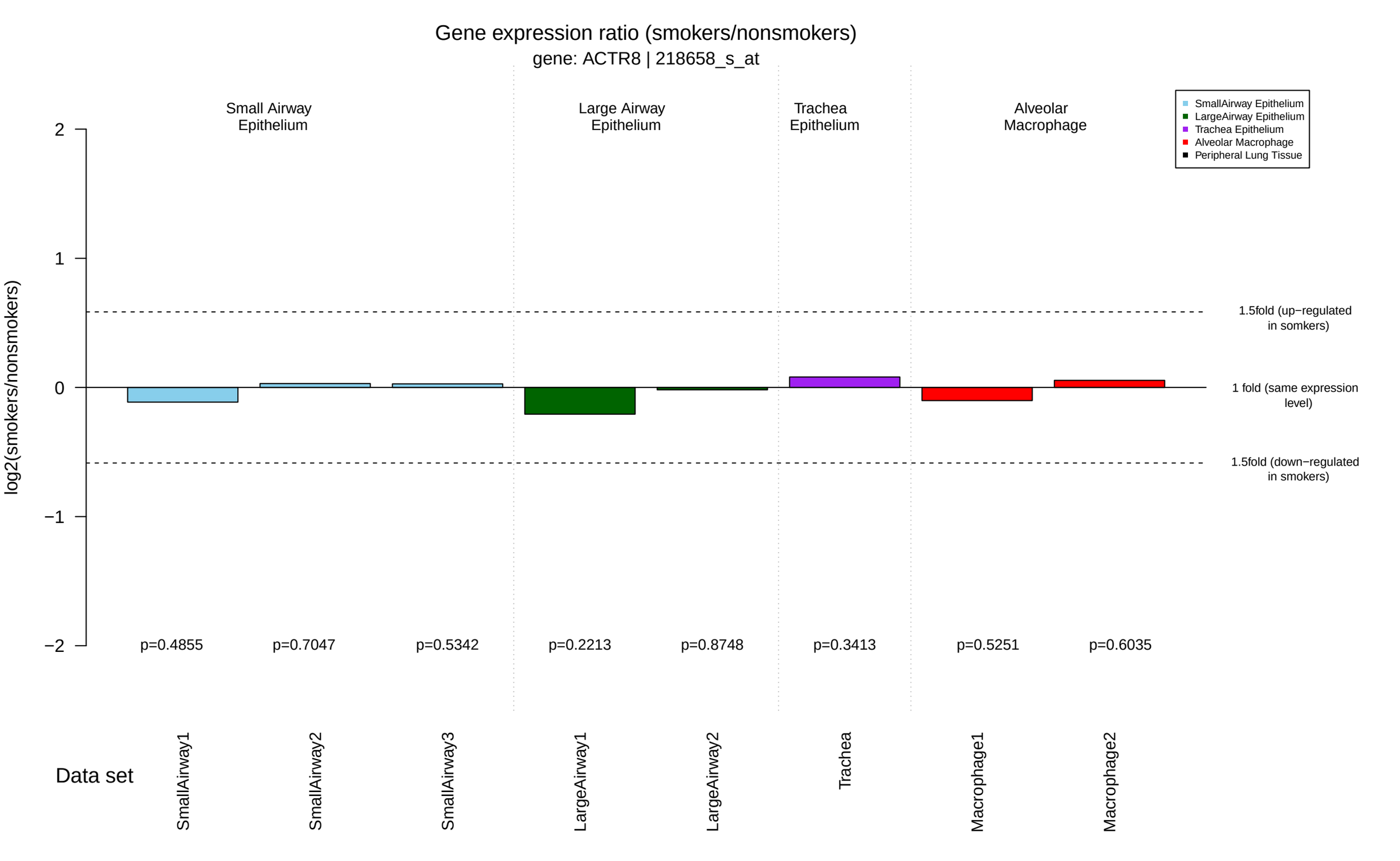
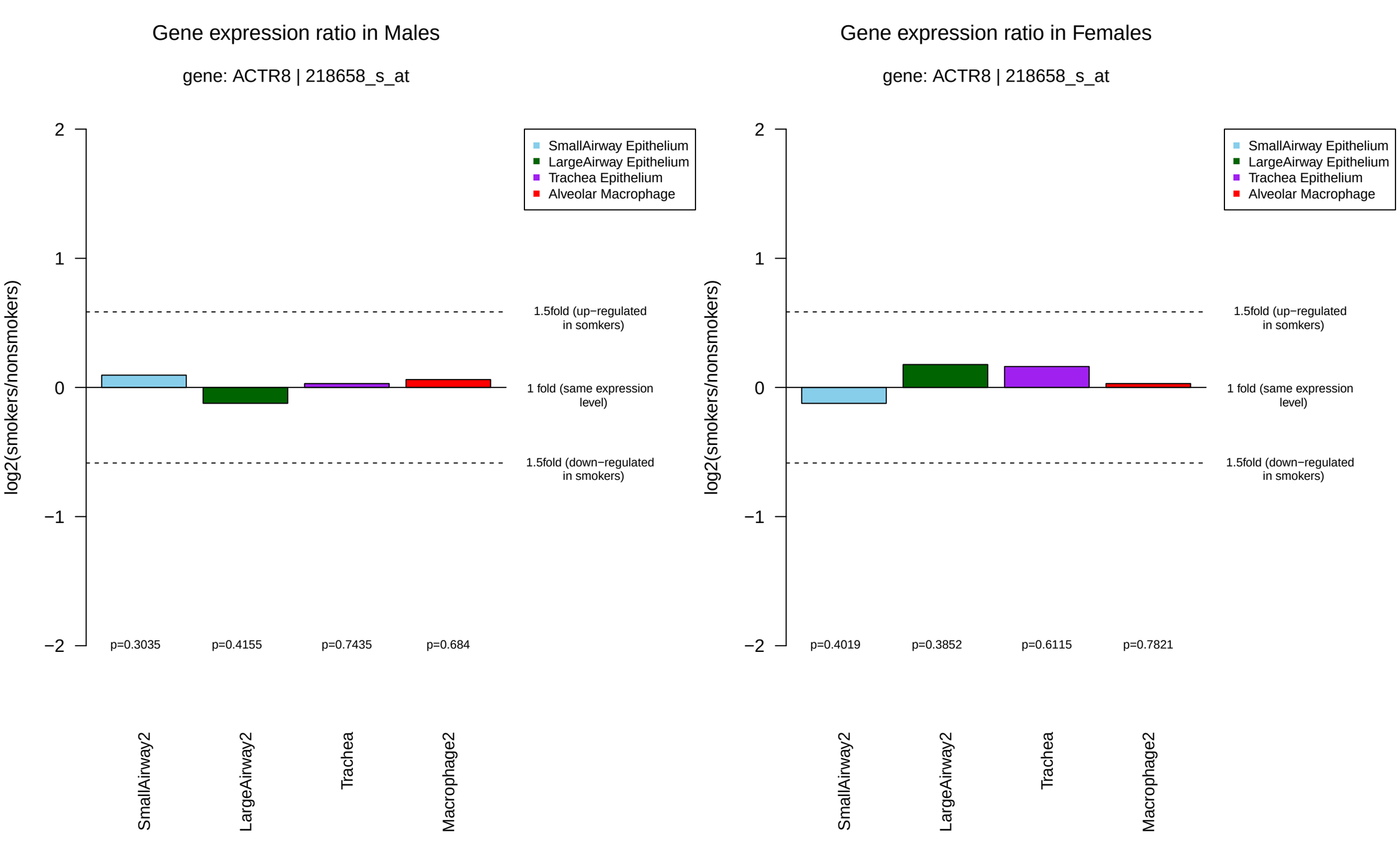
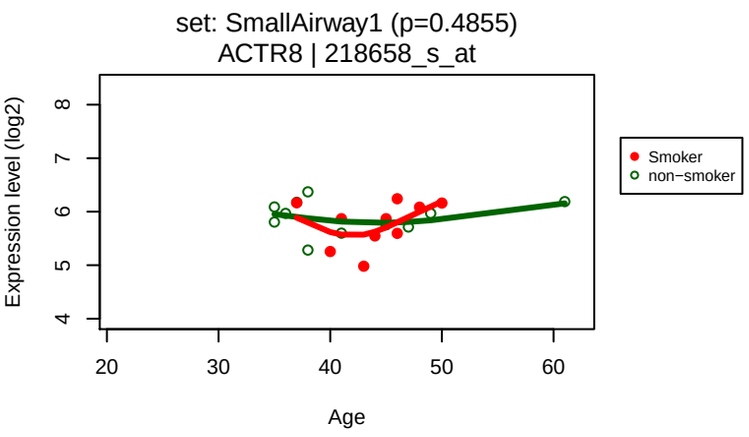
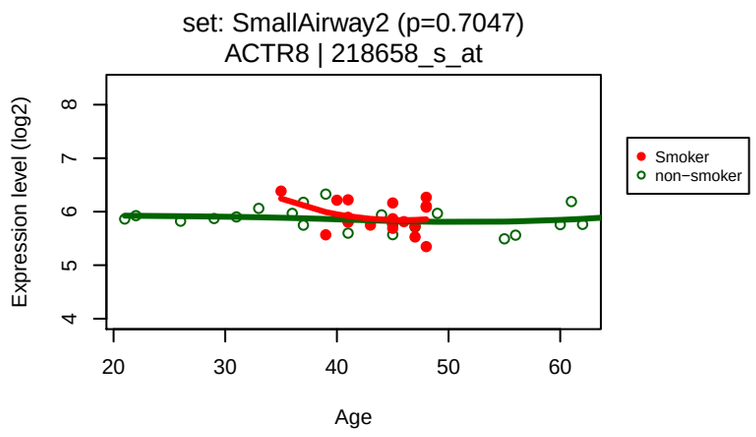
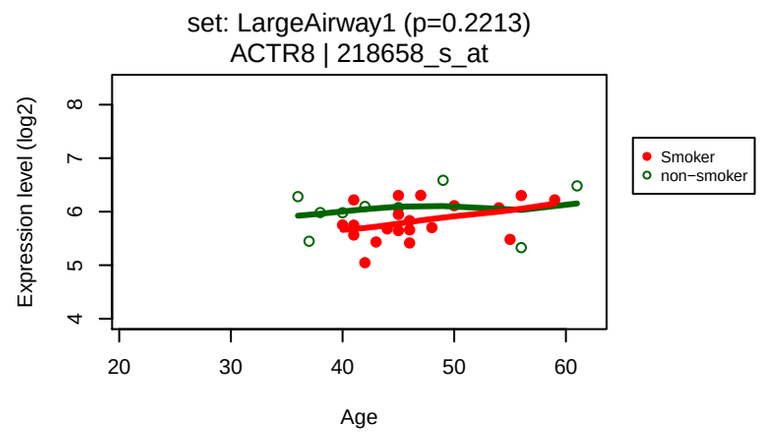

Gene expression of smokers and nonsmokers
gene: ACTR8 | 218658_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4855
p=0.7047
p=0.5342
p=0.2213
p=0.8748
p=0.3413
p=0.5251
p=0.6035
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

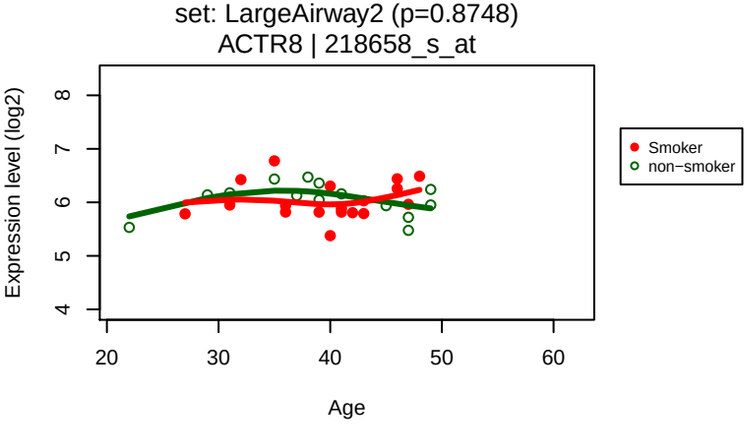

Gene expression ratio (smokers/nonsmokers)
gene: ACTR8 | 218658_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4855
p=0.7047
p=0.5342
p=0.2213
p=0.8748
p=0.3413
p=0.5251
p=0.6035
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

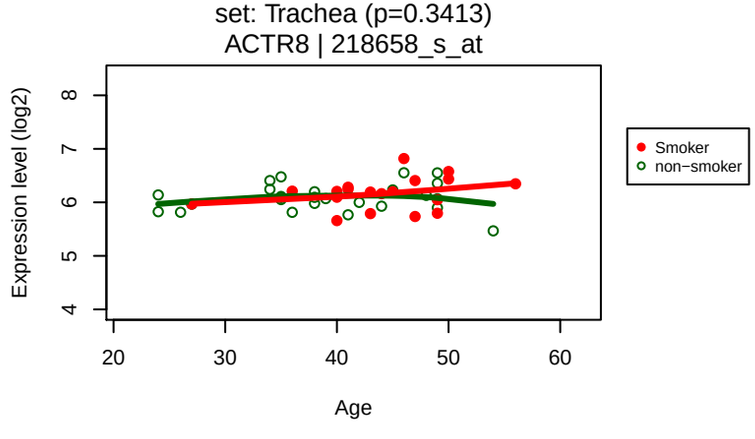

Gene expression ratio in Males
gene: ACTR8 | 218658_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3035
p=0.4155
p=0.7435
p=0.684
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ACTR8 | 218658_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4019
p=0.3852
p=0.6115
p=0.7821
SmallAirway2
LargeAirway2
Trachea
Macrophage2

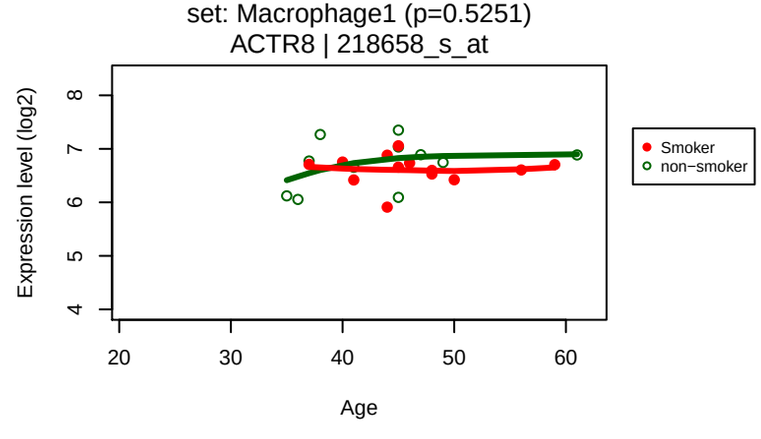

set: SmallAirway1 (p=0.4855)
ACTR8 | 218658_s_at
Expression level (log2)
Age
Smoker
non−smoker

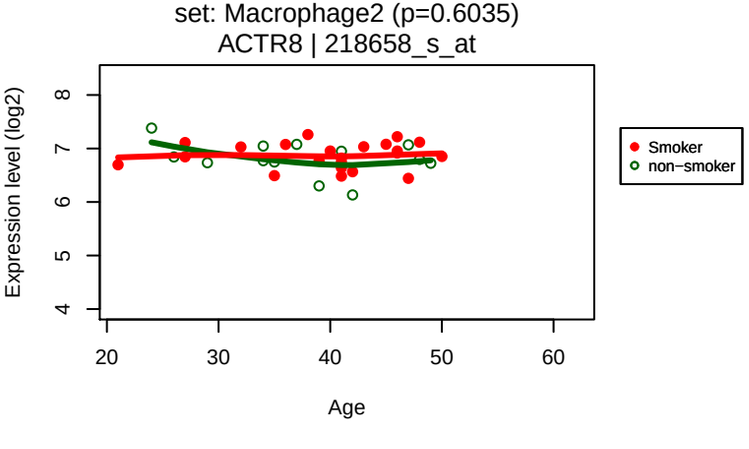

set: SmallAirway2 (p=0.7047)
ACTR8 | 218658_s_at
Expression level (log2)
Age
Smoker
non−smoker

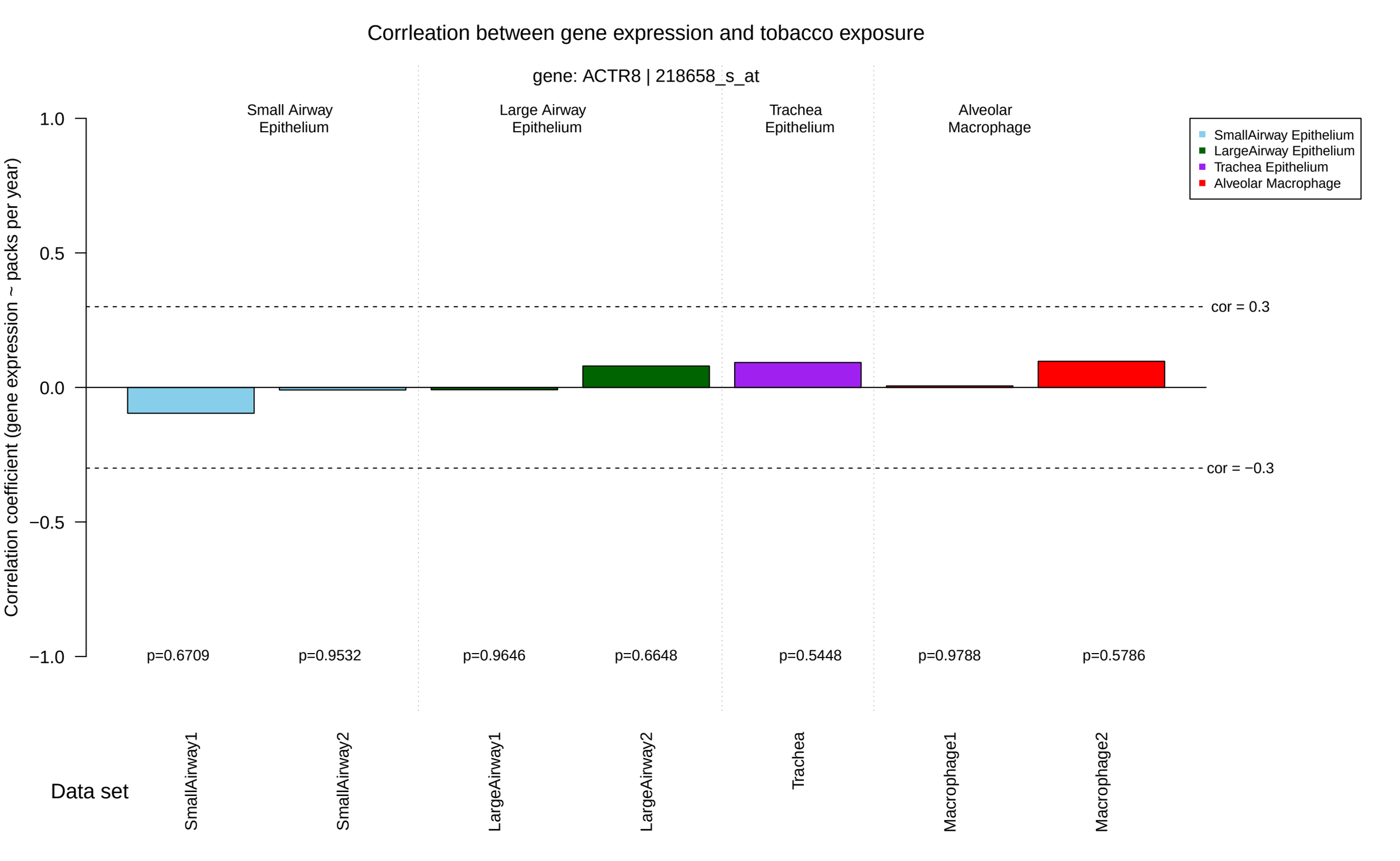

set: LargeAirway1 (p=0.2213)
ACTR8 | 218658_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8748)
ACTR8 | 218658_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3413)
ACTR8 | 218658_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5251)
ACTR8 | 218658_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6035)
ACTR8 | 218658_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ACTR8 | 218658_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6709
p=0.9532
p=0.9646
p=0.6648
p=0.5448
p=0.9788
p=0.5786
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2